Alessio Paradisi

Theoretical Physics · Nuclear Many-Body Theory

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I am a Master's student in theoretical physics at the University of Pisa, working on the microscopic description of dense neutron matter. My thesis takes a chiral nucleon–nucleon interaction through a self-consistent many-body calculation to ask whether neutron-star matter can become ferromagnetic. I enjoy the full path from formalism to working code, and I am looking for a PhD in nuclear theory, dense-matter astrophysics or computational many-body physics.

Education

2024 – 2026 **M.Sc. in Theoretical Physics** *University of Pisa*

Expected graduation 2026.

2021 – 2024 **B.Sc. in Physics** *University of Pisa*

Thesis: *Quantum Mechanics of a Particle on a Circle*, supervised by Prof. Claudio Bonati. Effect of non-trivial topology on the energy spectrum.

2016 – 2021 **Scientific High School Diploma** *Liceo Scientifico C. D'Ascanio, Montesilvano*

Master's Thesis

2025 – 2026 **Spin-Polarized Nuclear Matter Properties from Chiral Effective Field Theory** *University of Pisa*

Supervisor: Prof. Ignazio Bombaci Co-supervisor: Dr. Vishal Parmar

Brueckner–Hartree–Fock study of spin-polarized pure neutron matter with a local chiral potential including Δ-isobars, an interaction not previously applied to this system.

- Extended an inherited Fortran 77 BHF code to spin-polarized matter, rewriting most routines: separate spin populations, three spin-pair channels, and the corresponding statistical weights.
- Computed the equation of state, spin-resolved single-particle potentials and spin susceptibility up to twice saturation density, and decomposed the interaction energy by partial wave.
- Found no ferromagnetic instability in this range, and traced the result to the loss of the ${}^1S_0$ channel and the rise of ${}^3P_1$ with polarization and density.

Tools: Fortran 77, C, Python

Computational Projects

2026 **Thermodynamics of a free scalar field in 1+1 dimensions** *Numerical Lab III*

Lattice Monte Carlo with heatbath and microcanonical updates; energy density and trace anomaly across temperature scans and lattice refinements at fixed physical volume. C, Python

2025 **Topology of a quantum particle on a circle** *Numerical Lab II*

Path-integral Monte Carlo with local Metropolis updates in Euclidean time; results validated against analytic predictions at high and low temperature. C, Python

2025 **Critical properties of the 2D Ising model** *Numerical Lab I*

Single-cluster (Wolff) algorithm to beat critical slowing down; critical exponents from finite-size scaling. C, Python

2025 **Running couplings in QED and QCD** *Course projects*

Derived the QED running coupling and Uehling potential, and the one-loop QCD β function (asymptotic freedom), using dimensional regularization and the renormalization group.

Skills & Languages

Programming	Python, C, Fortran 77, Assembly
Tools	Linux (Ubuntu, Arch), Git, LaTeX, HTML
Methods	Many-body perturbation theory, Monte Carlo simulation, numerical linear algebra and quadrature
Languages	Italian (native) · English (fluent, Cambridge B2 First) · French (basic, A2)